

Esercizi capitolo 4

TRUE OR FALSE

False – Abstraction removes details to simplify a problem, not add them.

True – A metro map simplifies a real transport system while preserving key information.

False – Dynamic models show changes over time, while static models represent a system at a single point.

True – Abstraction focuses on relevant details and ignores unnecessary ones.

False – A directed graph can represent both static and dynamic models, depending on its use.

True – Leaky abstractions happen when hidden implementation details unexpectedly affect behavior.

False – A computer email system is an abstraction because it simplifies and models real-world communication.

True – Models simplify complex systems to aid understanding.

False – The main purpose of modeling is simplification, not replication of reality.

False – Accuracy refers to how close a measurement is to the true value, while precision refers to the level of detail.

MULTIPLE CHOICE QUESTIONS

b) To simplify a complex system by removing irrelevant details.

b) A photograph of a city (it captures full details rather than abstracting them).

a) A dynamic model changes over time, while a static model does not.

b) It can oversimplify and miss crucial details.

c) The concept of a vehicle (most generalized level).

b) To visualize and understand the relationships between entities.

c) Higher layers suppress more details and provide a broader view.

b) The core concepts or objects within a system.

c) Order teleported (not a realistic state).

a) They allow for efficient problem-solving by simplifying reality.

MULTIPLE MATCHING QUESTIONS

A – Abstraction: Representing essential details while ignoring irrelevant ones.

B – Static Model: Represents a system at a single point in time.

C – Dynamic Model: Shows how a system changes over time.

D – Entity: A core concept or object within a system.

E – Relationship: A connection that links entities within a system.

F – Layers of Abstraction: Structuring concepts at different levels of detail.

G – Leaky Abstraction: When hidden details unexpectedly affect a system.

H – Modelling: Creating representations to analyze and solve problems.

I – Precision: The level of refinement in measurement.

J – Accuracy: The closeness of a measurement to the true value.

COMPREHENSION QUESTIONS FOR CLASSROOM DISCUSSION

Why is abstraction important in computational thinking?

It simplifies complex systems by focusing on relevant details, making problem-solving more efficient.

How does a metro map serve as an abstraction?

It represents routes and stations without unnecessary geographical details, helping users navigate efficiently.

What is the main difference between static and dynamic models?

A static model represents a system at a single point in time, while a dynamic model shows changes over time.

How can leaky abstractions impact a system?

They introduce unintended behavior when hidden details interfere with expected functionality.

Why do models not always represent reality with perfect accuracy?

Models simplify reality by omitting details, which can lead to approximations rather than exact representations.

What role do layers of abstraction play in problem-solving?

They allow for problem decomposition by breaking complex systems into manageable parts.

How can modelling help in understanding complex systems?

It provides visual and logical representations, making it easier to analyze and predict system behavior.

Give an example of a real-world situation where abstraction is useful.

Traffic navigation apps simplify real-world maps by highlighting only roads, routes, and key locations.

Why is precision different from accuracy in computational models?

Precision refers to the level of detail, while accuracy is how close a value is to the true measurement.

How does a state machine diagram represent different system states?

It uses states and transitions to depict how a system progresses from one stage to another based on specific events.